

Econometrics II (JEB110)

Home Assignment 2

Deadline: November 10, 2025, 11:59 PM

Dear students,

You are encouraged to work in groups of two people. In that case hand only one solution with names of both participants on the first page. In case you cannot find any colleague to form a group with, you can write me an email and I will match you with somebody. Submit your solution electronically via SIS. Pack all electronically submitted files into a zipped folder named "**Surname1_Surname2_HA1**".

Solution of the first problem should be handwritten and might be submitted on paper to the mailbox No.14 (Barbara Pertold-Gebicka) or scanned and submitted electronically. Solution of the second problem can be written in any text-editing software of your choice (e.g., MS Word, LATEX) and should be submitted electronically. Attach both R "log" file with results and "rmd" file with the programming code to your assignment. If using another statistical software, attach the corresponding file(s). Use comments in your programming code to describe in few words what each command means. In case of problems with R, consult the official online manual and help module.

Keep in mind that not only the correctness of the results is assessed, but also the text editing quality is taken into consideration. Answer only what you are asked for.

Note that the use of **Artificial Intelligence (AI)** tools is only allowed to improve your writing style and grammar or to generate inspiration. However, treat the AI tools' output carefully and with caution, respecting the principles of academic integrity and the requirement to produce your own original work that is based on reliable and verifiable sources of information. Simple copying of the text generated by the AI tools is not permitted. Consult the full text of AI guidelines [here](#).

Problem 1

Consider the standard two-period, two-group difference-in-differences (DiD) setup:

$$y_{it} = \beta_0 + \beta_1 T_t + \beta_2 D_i + \beta_3 T_t \cdot D_i + u_{it}. \quad (1)$$

where y_{it} is the outcome for unit i at time t , $D_i = 1$ for the treatment group and zero otherwise, $T_t = 1$ for the period after the policy change and zero otherwise, and u_{it} is the unobserved disturbance. β_3 is the effect of the policy change.

- (a) Explain what is meant by the parallel trend assumption.
- (b) Suppose that the parallel trends assumption fails because for units belonging to the control group $E[y_{it}|T_t] = (\beta_1 + \gamma) \cdot T_t$. Derive the expected value of the DiD estimator of the effect of policy change in this case if Equation (1) is used for estimation.
- (c) Based on the above derivations, discuss how the sign of the bias in the DiD estimator of the policy change effect depends on the sign of γ , i.e., whether the trend in the control group is steeper or flatter than the trend in the treatment group.

Problem 2

The purpose of this assignment is to practice working with panel data within the scope of a bachelor's program. When approaching Problem 2, use panel data methods introduced in Econometrics II (JEB110). Present your work in a form of a short (maximum 5-page) report that will summarize your empirical approach, the data used, results and comments/conclusions. Attach both the R“log” files with results and “rmd” file with the programming commands. In case you use another statistical software, attach the respective relevant files. Treat the points below as a suggestive checklist, but please, do not write your report as a list of ten points.

Your task is to estimate the relationship between the length of the paid parental leave and the extent of the gender pay gap (difference between average gross hourly earnings of male and female employees as % of male gross earnings) across European countries. There is an ongoing discussion about the trade-off between the time mothers spend out of the labor market after childbirth and their future earnings. Let us contribute to this discussion by analyzing a brand new database of parental leave policies across 21 European countries. An extract from this database is available in SIS for your use. Note that the data is on the country level, so you cannot estimate the gender pay gap in the way we estimated it during the lecture. For the purpose of this assignment, you will take the values of the unadjusted gender pay gap estimated by the Eurostat.

Make sure that you understand the data, know what you are going to estimate, that you include the relevant explanatory variables, and that you interpret your estimates accordingly. If you do not trust your estimates, write why.

1. Think about the relationship that you are going to estimate. Which factors and why might affect the gender pay gap? Is this relationship expected to be linear? Build a regression equation based on this discussion.
2. Download data for the gender pay gap and, potentially, other relevant variables from Eurostat database for the years 2023, 2018, and 2013.
3. Prepare the dependent and explanatory variables. In which units are

they measured? Maybe some of the variables should be in logs? Explain your choices.

4. Write down the equation that you are going to estimate. Do not forget to use correct indices and explain the meaning of each variable used in the model (i.e., what each variable means, in which units it is measured, etc.). Prepare summary statistics of the data.
5. Estimate your model using the data for 2023. Interpret the results and discuss whether they can be trusted.
6. Next, consider all three years of the data. How can this help you?
7. Build a panel data model that will be used to estimate the relationship between the length of the paid parental leave and the extent of the gender wage gap. Discuss whether you expect the fixed effect (a_i) to be correlated with any of the explanatory variables. Which estimation technique for panel data would you use to estimate your model? Why?
8. Do not forget to discuss whether all relevant assumptions are satisfied. You might use theoretical/intuitive arguments and/or formal tests to support your claims.
9. Estimate your model using one or several approaches of your choice. Report results in a comprehensive table. If you report more than one set of results (from more than one specification and/or more than one estimation technique), then comment on which of them is the most reliable and why.
10. Comment on your estimates. Do you find them intuitive? Is there something you would do differently if you had more time and/or better data? Why?